Introduction to Symmetric Pairs

—from the perspective of $K\backslash G/B$

Rui Xiong

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Symmetric Pairs

Real Lie algebras

Over complex numbers $\mathbb{C}$, we have

$$\left\{\begin{matrix}\text{semisimple}\\ \text{Lie algebras}\end{matrix}\right\} = \{\text{Dynkin diagrams}\}.$$

How about real numbers $\mathbb{R}$?

We will explain that over real numbers $\mathbb{R}$

$$\left\{\begin{matrix}\text{semisimple}\\ \text{Lie algebras}\end{matrix}\right\} \approx \left\{\begin{matrix}\text{Dynkin diagrams}\\ \text{with involutions}\end{matrix}\right\}.$$

The symbol "$\approx$" will be made precise.

Involutions

General theory of Galois descent tells that

$$\{\text{real forms of } \mathfrak{g}\} = \left\{ \begin{array}{c} \text{anti-linear involutions} \\ \sigma : \mathfrak{g} \to \mathfrak{g}, \text{ i.e. } \sigma^2 = \mathrm{id}, \\ \sigma(ax + y) = \bar{a}\sigma(x) + \sigma(y) \end{array} \right\}.$$

It can be further classified by Galois cohomology

$$H^1(Gal(\mathbb{C}/\mathbb{R}), \mathrm{Aut}(\mathfrak{g}))$$

but this is not today's theme.

Split and Compact Forms

There is two distinguished elements of $\{\text{real forms of } \mathfrak{g}\}$.

The split real form:

$$\sigma : \begin{aligned} E_i &\mapsto E_i \\ F_i &\mapsto F_i \\ H_i &\mapsto H_i \end{aligned}$$

Example: $GL_n \ni A \mapsto \overline{A}$.

The compact real form:

$$\sigma : \begin{aligned} E_i &\mapsto -F_i \\ F_i &\mapsto -E_i \\ H_i &\mapsto -H_i \end{aligned}$$

Example: $GL_n \ni A \mapsto A^H$.

Note that the involutions above are all anti-linear.

Symmetric Pairs

A nontrivial fact of Cartan says,

$$\left\{\begin{array}{c}\text{anti-linear involutions}\\ \sigma:\mathfrak{g}\to\mathfrak{g},\ \text{i.e. } \sigma^2=\mathrm{id},\\ \sigma(ax+y)=\bar{a}\sigma(x)+\sigma(y)\end{array}\right\}\xleftrightarrow{1:1}\left\{\begin{array}{c}\text{linear involutions}\\ \tau:\mathfrak{g}\to\mathfrak{g},\ \text{i.e. } \tau^2=\mathrm{id},\\ \tau(ax+y)=a\tau(x)+\tau(y)\end{array}\right\}.$$

Precisely, for any $\tau\in RHS$, there exists a compact involution θ such that $\tau\theta=\theta\tau\in LHS$.

So the question reduces to classify linear involutions. We will call

$$(G, K=G^\tau) \text{ or } (\mathfrak{g}, \mathfrak{k}=\mathfrak{g}^\tau)$$

a **symmetric pair**.

Type A

When $G = GL_n$, we have three types

type	involution	$K \subset G$	$G_{\mathbb{R}}$
AI	$g \mapsto (g^t)^{-1}$	$O_n \subset GL_n$	$GL_n(\mathbb{R})$
AII	$g \mapsto J_{2m}(g^t)^{-1}J_{2m}$	$Sp_{2m} \subset GL_{2m}$	$GL_m(\mathbb{H})$
AIII	$g \mapsto I_{p,q} \cdot g \cdot I_{p,q}^{-1}$	$GL_p \times GL_q \subset GL_{p+q}$	$U_{p,q}$

where $J_{2m} = \begin{bmatrix} & I_m \\ -I_m & \end{bmatrix}$ and $I_{p,q} = \begin{bmatrix} I_p & \\ & -I_q \end{bmatrix}$.

Classifications

Good news: **The symmetric pairs are classified**;

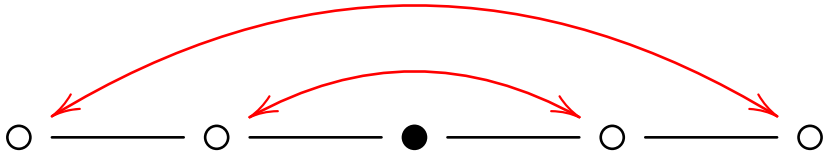
Bad news: **there are two classifications**.

We have

$$\left\{\begin{matrix}\mathbb{R}\text{eal forms of} \\ \text{semisimple} \\ \text{Lie algebras}\end{matrix}\right\} \begin{matrix} = \{\text{Satake diagrams}\} \\ \\ = \{\text{Vogan diagrams}\}\,. \end{matrix}$$

They are both given by an involution on the Dynkin diagram colored by black and white.

Comparison



Satake diagram

$$\begin{bmatrix} E_i \mapsto -F_{\tau(i)} \\ F_i \mapsto -E_{\tau(i)} \\ H_i \mapsto -H_{\tau(i)} \end{bmatrix} \circ \mathrm{Ad}_{w_\bullet}$$

where $w_\bullet$ is the longest Weyl group element of black nodes.

Example: $(GL_6, GL_2 \times GL_4)$.

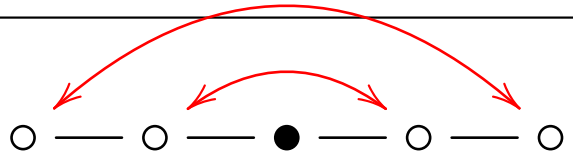
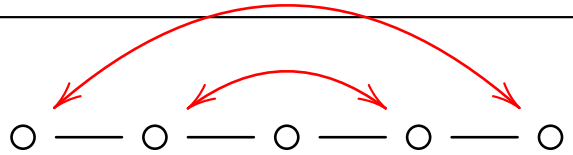
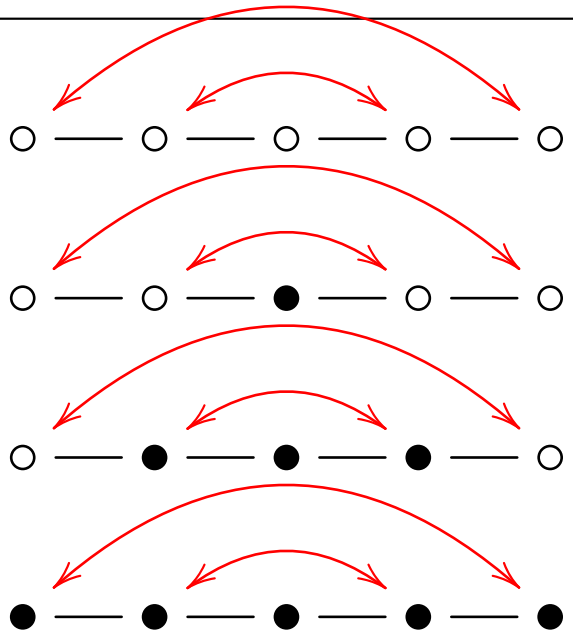
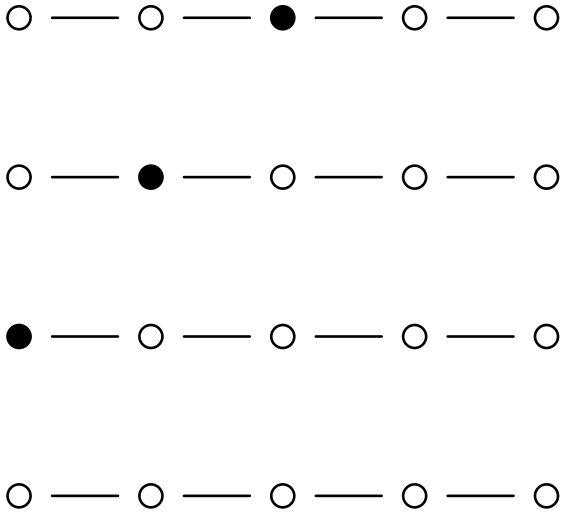
Vogan diagram

$$\begin{bmatrix} E_i \mapsto \pm E_{\tau(i)} \\ F_i \mapsto \pm F_{\tau(i)} \\ H_i \mapsto H_{\tau(i)} \end{bmatrix}$$

where $\pm = +$ if i is white, and $\pm = -$ if i is black.

Example: (GL_6, O_6).

Example: Type A

type	Satake	Vogan
AI		
AII		
AIII		

Remarks on well-definedness

To make a Satake diagram well-defined, we have to assume

$$\tau|_{\bullet} = -w_{\bullet}$$

i.e. the restriction of the involution on black nodes are induced by $-w_{\bullet}$.

In Vogan diagram, different coloring could give the same symmetric pair, e.g.

$$\bullet - \circ \iff \bullet - \bullet \iff \circ - \bullet .$$

But we can always assume non-fixed nodes are white and at most one node is black.

Family of Examples — Split and compact

Split case (G, K).

- all nodes of the Satake diagram are white; the involution is trivial;
- all nodes of a Vogan diagram are black; the involution is $-w_0$ (non-standard).

Compact case $(G, K = G)$.

- all nodes of the Satake diagram are black; the involution is $-w_0$;
- all nodes of a Vogan diagram are white; the involution is trivial.

Family of Examples — Quasi-split

We say (G, K) is **quasi-split** if the Satake diagram contains only white nodes.

- all nodes of the Satake diagram are white;
- all nodes of a Vogan diagram are black.

The involution differ by $-w_0$.

Equivalently, there exists a θ-split Borel B, i.e.

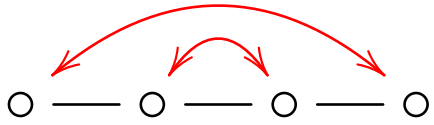
$$B \cap \theta(B) \text{ is a maximal torus.}$$

Note that the standard Borel from the Vogan diagram is θ-invariant, i.e.

$$B = \theta(B).$$

Example

Satake diagram

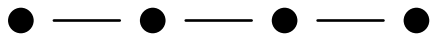


The involution is given by

$$GL_5 \ni [a_{ij}] \mapsto [a_{6-i,6-j}].$$

$$\Rightarrow K = GL_2 \times GL_3$$

Vogan diagram

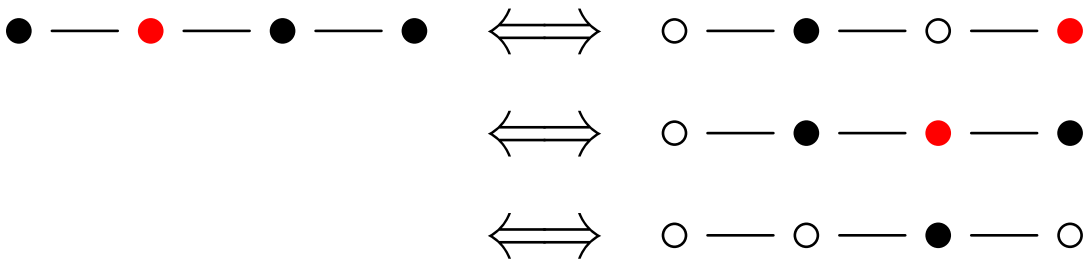


The involution is given by

$$GL_5 \ni [a_{ij}] \mapsto [(-1)^{i+j} a_{ij}].$$

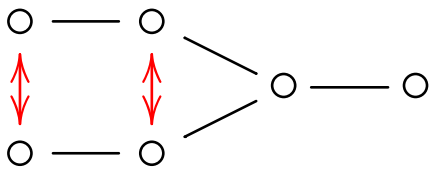
$$\Rightarrow K = GL_2 \times GL_3$$

We can also play the following trick on Vogan diagram:

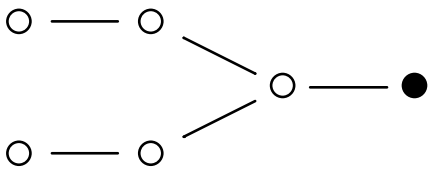


Example

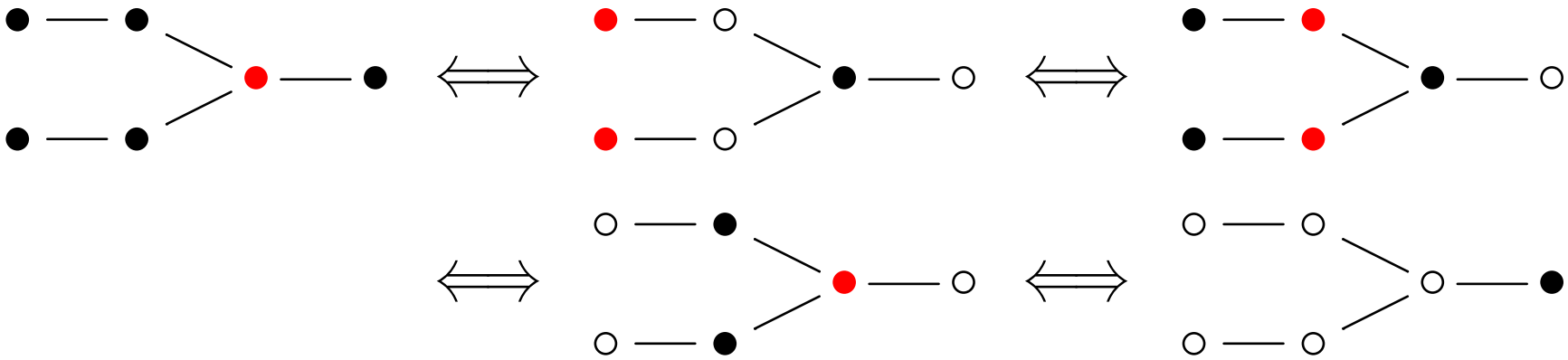
Satake diagram



Vogan diagram



Here is the proof:

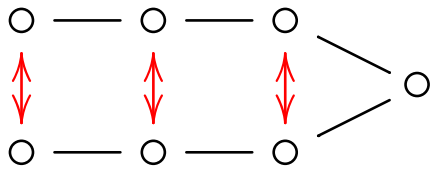


Family of Examples — Folding

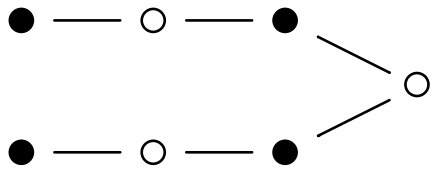
The degree-2 folding gives a symmetric pair (G, K).

- all nodes of a Vogan diagram is white; the involution is the folding.

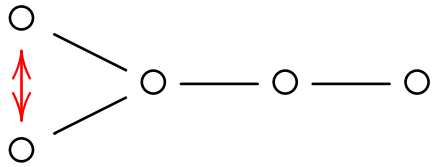
Vogan diagram



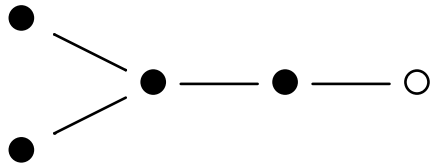
Satake diagram



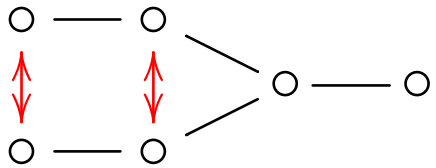
Vogan diagram



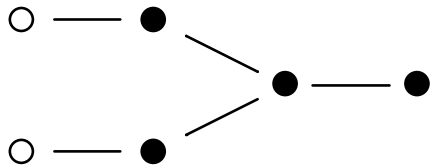
Satake diagram



Vogan diagram



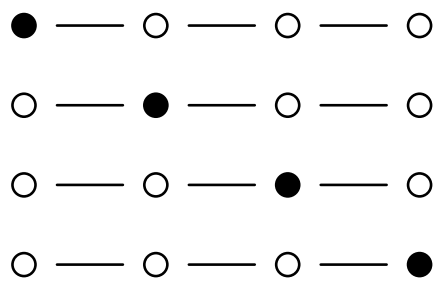
Satake diagram



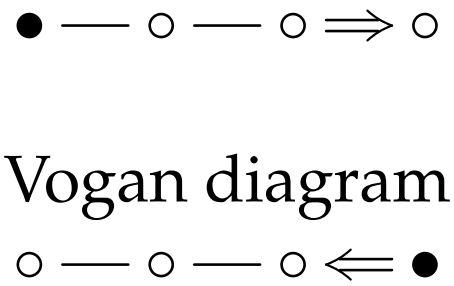
Family of Example — Hermitian

We say (G, K) is **Hermitian** if K is a Levi subgroup.

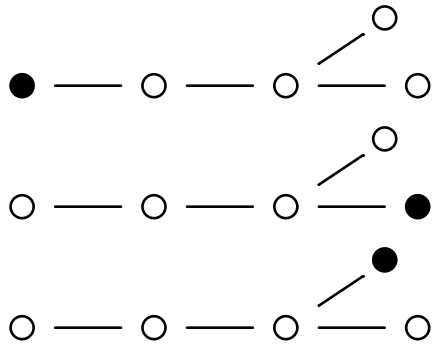
Vogan diagram



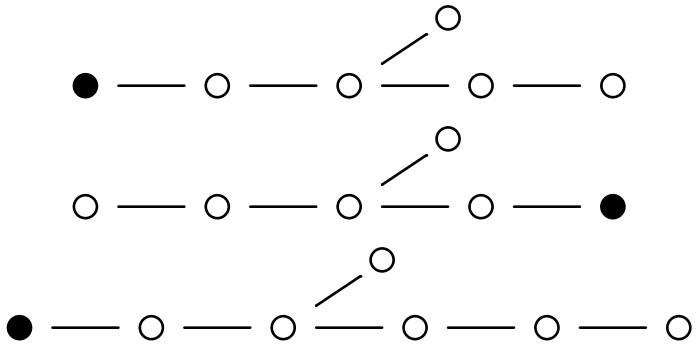
Vogan diagram



Vogan diagram



Vogan diagram



Classical Types

type	$K \subset G$
AI	$O_n \subset GL_n$
AII	$Sp_{2n} \subset GL_{2n}$
AIII	$GL_p \times GL_q \subset GL_{p+q}$

type	$K \subset G$
BDI	$O_p \times O_q \subset O_{p+q}$
DIII	$GL_n \subset SO_{2n}$

type	$K \subset G$
CII	$Sp_p \times Sp_q \subset Sp_{p+q}$
CI	$GL_n \subset Sp_{2n}$

$K\backslash G/B$

SYMMETRIC SPACES, FLAG VARIETIES, ETC

Symmetric Spaces

A symmetric pair (G, K) defines a symmetric space G/K, or its compact form G_c/K_c. The classical types have the following identification (up to homotopy):

$$
\begin{aligned}
(G \times G)/G &= G \\
GL_{p+q}/(GL_p \times GL_q) &\approx \text{complex Grassmannian} \\
O_{p+q}/(O_p \times O_q) &\approx \text{real Grassmannian} \\
Sp_{p+q}/(Sp_p \times Sp_q) &\approx \text{quaternion Grassmannian} \\
GL_n/O_n &= \text{symmetric invertible matrices} \\
GL_n/Sp_n &= \text{anti-symmetric invertible matrices} \\
SO_{2n}/GL_n &\approx \text{isotropic Grassmannian} \\
Sp_{2n}/GL_n &\approx \text{Lagrangian Grassmannian}
\end{aligned}
$$

NB: (G, K) is Hermitian $\Leftrightarrow$ G_c/K_c is a partial flag variety.

Infinite types

Many objects are classifying space or loop space when the index $\to \infty$.

$$\begin{aligned} GL/(GL \times GL) &= Gr_{\mathbb{C}} = BGL_\infty = \Omega GL_\infty \\ O/(O \times O) &= Gr_{\mathbb{R}} = BO_\infty \\ Sp/(Sp \times Sp) &= Gr_{\mathbb{H}} = BSp_\infty \\ SO/GL &= IGr_\infty = \Omega O_\infty \\ Sp/GL &= SGr_\infty = \Omega Sp_\infty. \end{aligned}$$

Their cohomology rings

$$\begin{aligned} H^*(BGL_\infty) = H^*(\Omega GL_\infty) &= \mathbb{Q}[p_1, p_2, p_3, \ldots] = \Lambda \\ H^*(BO_\infty) = H^*(BSp_\infty) &= \mathbb{Q}[p_2, p_4, p_6, \ldots] \subset \Lambda \\ H^*(\Omega O_\infty) = H^*(\Omega Sp_\infty) &= \mathbb{Q}[p_1, p_3, p_5, \ldots] \subset \Lambda. \end{aligned}$$

Infinite types and Bott periodicity

CII $Sp \times Sp \subset Sp$	AII $Sp \subset GL$	DIII $GL \subset O$
C $Sp \subset Sp \times Sp$	$\longrightarrow$ apply Ω	BD $O \subset O \times O$
CI $GL \subset Sp$	AI $O \subset GL$	BDI $O \times O \subset O$

A $GL \subset GL \times GL$
AIII $GL \times GL \subset GL$

Spherical Subgroups

A subgroup H of G is called **spherical** if the set

$$H\backslash G/B = \{H\text{-orbits of } gB \in G/B\}$$

is finite, where G/B is the flag variety.

The Borel subgroup B is spherical. We have **Bruhat decomposition**

$$B\backslash G/B \overset{1:1}{=\!=} \text{Weyl group.}$$

Each orbit is known as a **Schubert cell**.

The group K in a symmetric pair is spherical (a version of **Iwasawa decomposition**).

$$K\backslash G/B \overset{1:1}{=\!=} \text{a finite set.}$$

We are interested in the K-orbits of G/B.

Open and closed orbits

There is a unique open orbit. Actually, this is given by the **Satake diagram**. It is not hard to see

$$\mathrm{Lie}\, K + \mathrm{Lie}\, B = \mathrm{Lie}\, G$$

under the Satake involution. Equivalently, KB is dense in G (a version of Iwasawa decomposition).

There are finitely many closed orbits. Actually, these are all from the **Vogan diagrams**. Since $K \cap B$ is a Borel of K, the orbit of $1B$ is isomorphic to

$$K/K \cap B = \text{flag variety of } K$$

thus it is closed.

Examples

When $G = GL_n$, the flag variety

$$G/B = \{0 = V_0 \subset V_1 \subset \cdots \subset V_{n-1} \subset V_n = \mathbb{C}^n\}.$$

type	$K \subset G$	index set of $K \backslash G/B$
AI	$O_n \subset GL_n$	$\{w \in S_n : w^2 = 1\}$
AII	$Sp_{2m} \subset GL_{2m}$	$\{w \in S_{2m} : w^2 = 1, w(i) \neq i\}$
AIII	$GL_p \times GL_q \subset GL_{p+q}$	$\{(p,q)\text{-clans}\}$

Other classical types are given by clans with symmetry.

Examples

Note that

> *For any invertible symmetric matrix A, there exists an invertible matrix U such that $UAU^T = I$.*

We have

$$GL_n/O_n = \{\text{symmetric inverstible matrices}\}.$$

The classification is B-orbit of GL_n/O_n is the following exercise in linear algebra.

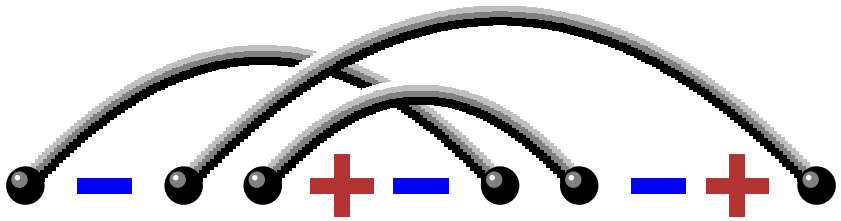
> *For any invertible symmetric matrix A, there exists an upper triangular matrix U such that UAU^T is a permutation matrix.*

Clans

A (p, q)-**clan** is a partial matching of $p + q$ nodes with unmatched nodes colored by $+$ or $-$ with

$$\#\{+\} - \#\{-\} = p - q.$$

For example, the following is a $(5, 6)$-clan

$\gamma =$ 

$$\dot{\gamma} = \overset{e_1}{\underset{e_8}{+}}\ e_6\ \overset{e_2}{\underset{e_{11}}{+}}\ \overset{e_3}{\underset{e_9}{+}}\ e_4\ e_7\ e_8\ e_9\ e_{10}\ e_5\ e_{11},$$

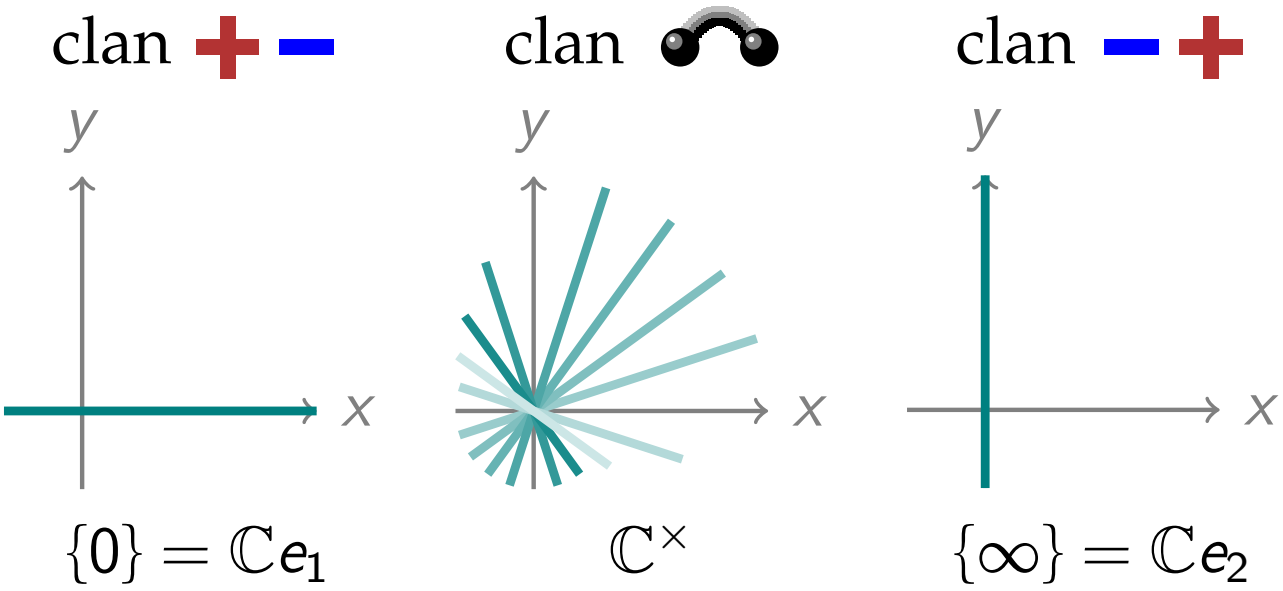
The orbit is $Y_\gamma^\circ := K\dot{\gamma}B/B$.

Example

Consider the case

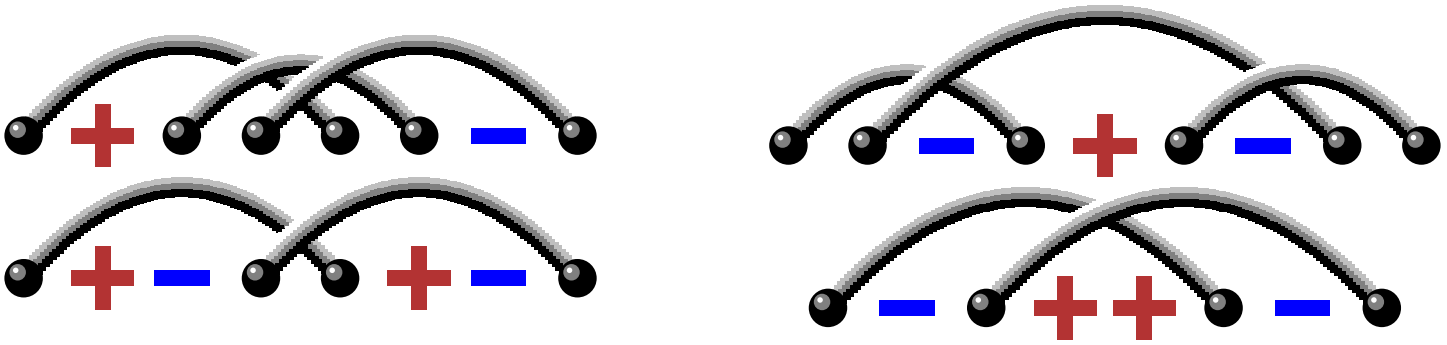
$$GL_1 \times GL_1 \subset GL_2 \circlearrowright \mathbb{P}^1 = \{0 \subset L \subset \mathbb{C}^2\} = \mathbb{C} \sqcup \{\infty\}.$$

There are three orbits



Classical types

type	$K \subset G$	description
CI	$GL_n \subset Sp_{2n}$	{skew symmetric clans}
DIII	$GL_n \subset O_{2n}$	{skew symmetric clans without invariant mathcing}
BDI	$O_p \times O_q \subset O_{p+q}$	{symmetric clans}
CII	$Sp_{2p} \times Sp_{2q} \subset Sp_{2p+2q}$	{symmetric clans without invariant mathcing}



Family of examples — Folding

In general, for folding type

$$K\backslash G/B \overset{1:1}{\longleftrightarrow} W_K\backslash W.$$

Note that W_K is obtained by foling.

For example, $Sp_{2n} \subset GL_{2n}$,

$$\left\{ w \in S_{2n} : \begin{matrix} w^2 = 1, \\ w(i) \neq i \end{matrix} \right\} \begin{matrix} = \{ g w_0 g^{-1} : g \in S_{2n} \} \\ \cong C_n \backslash S_{2n} \end{matrix}$$

where C_n is the Weyl group of Sp_n.

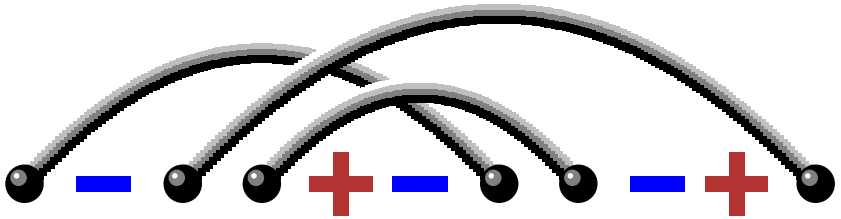
Family of examples — Hermitian

Assume $(G, K = P \cap P^{-})$ is a Hermitian symmetric pair. Then

$$K\backslash G/B \xleftrightarrow{1:1} \left\{ (w, \beta_1, \ldots, \beta_r) : \begin{array}{c} w \in W^P, \beta_1, \ldots, \beta_r \in \mathrm{Inv}(w) \\ \text{mutually orthogonal roots} \end{array} \right\}.$$

For example, $GL_p \times GL_q \subset GL_{p+q}$,

$\gamma =$



$$w^{-1} = 1 \;\; 6 \;\; 2 \;\; 3 \;\; 4 \;\; 7 \;\; 8 \;\; 9 \;\; 10 \;\; 5 \;\; 11.$$

The roots β's are given by the matchings.

Hecke algebra actions

There is a monoid structure over the Weyl group

$$s_i * s_i = s_i \qquad \text{braid relations.}$$

This monoid acts on $K\backslash G/B$. Geometrically,

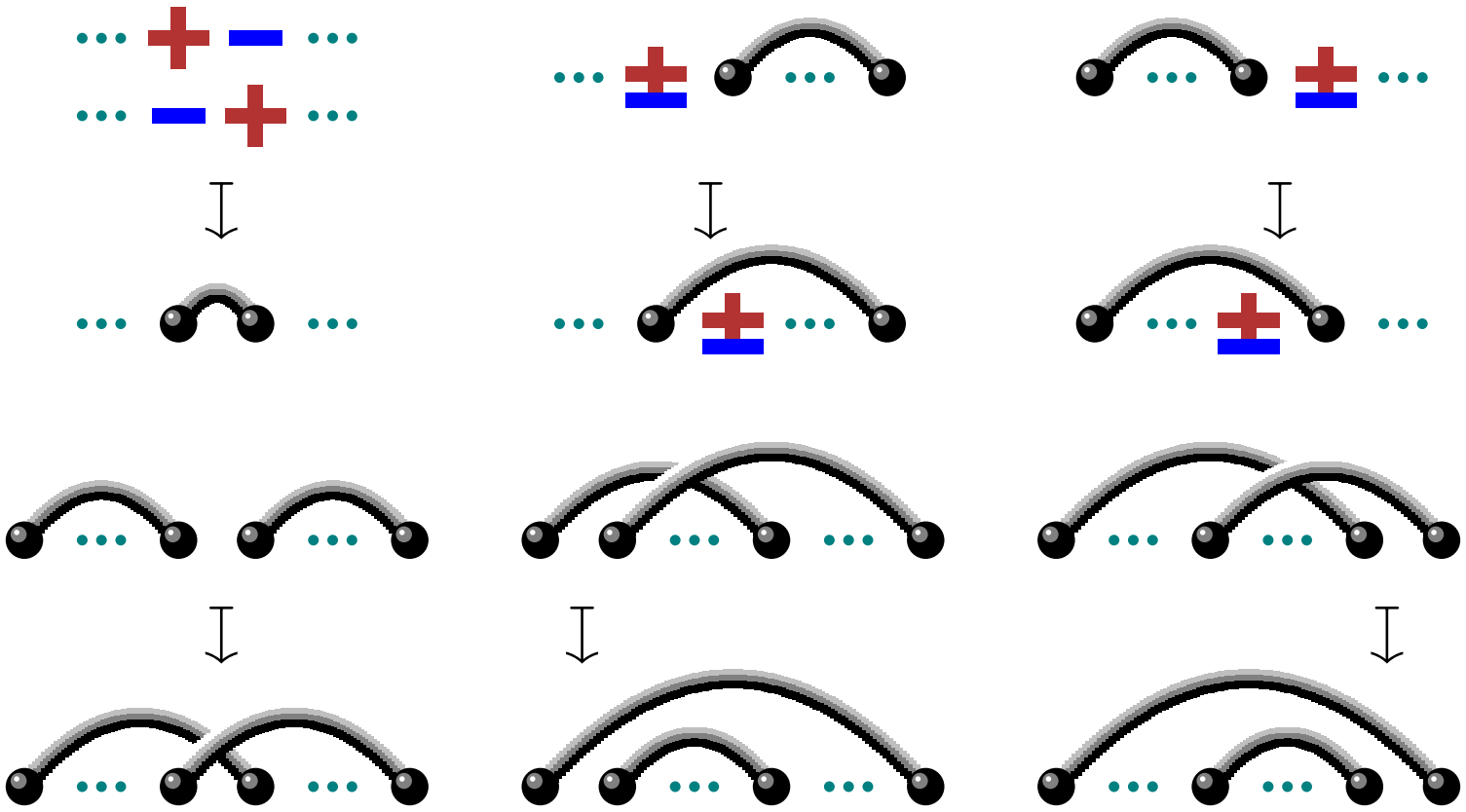
$$\text{orbit } O \xmapsto{s_i *} \text{the orbit which is dense in } \pi^{-1}(\pi(O))$$

for $\pi : G/B \to G/P$.

By a similar way, there is a Hecke algebra action (with certain local system introduced), known as **Lusztig–Vogan module**.

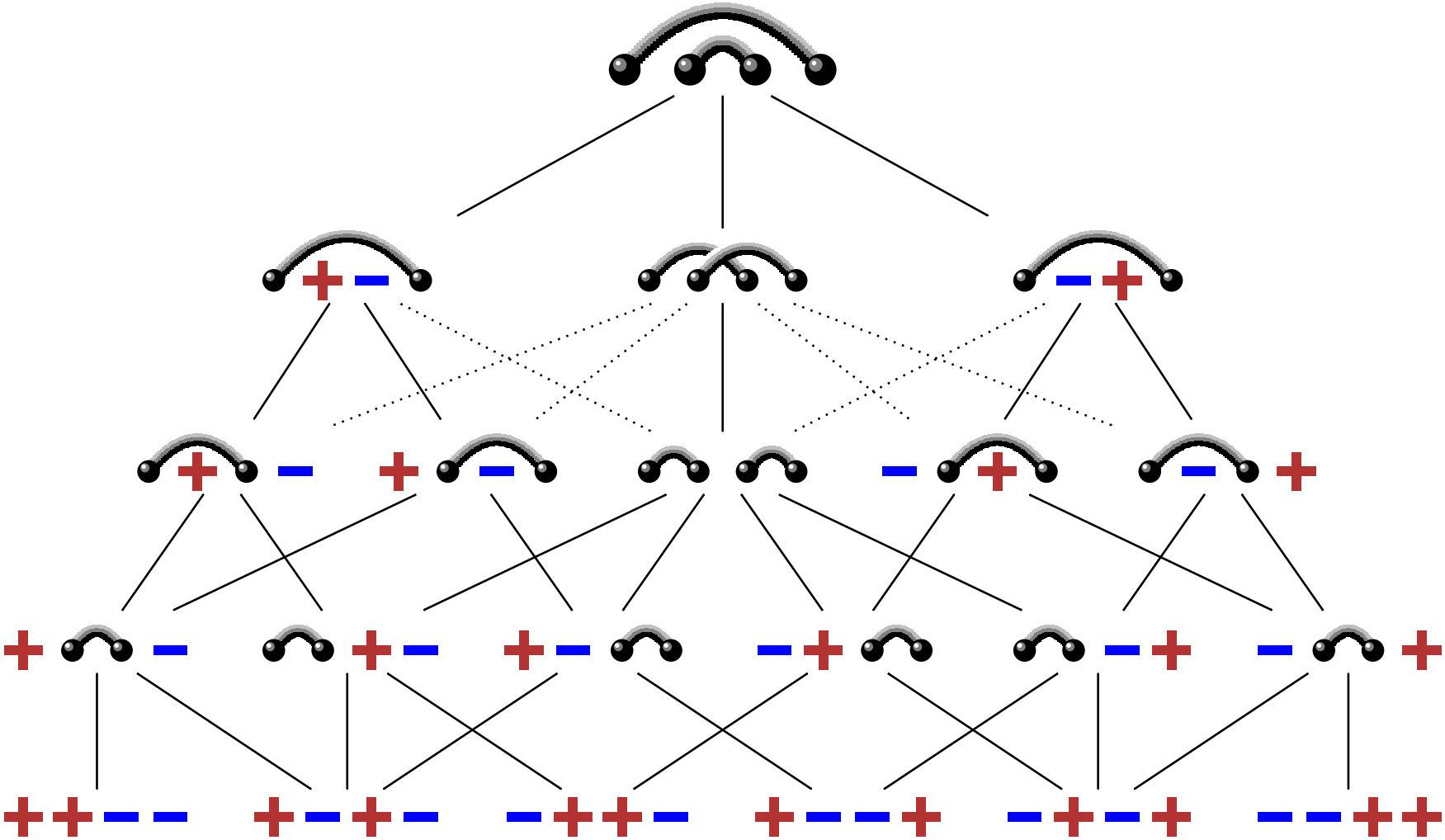
Orders on Clans

The weak order of clans are generated by



There is also a strong order induced by orbital closure.

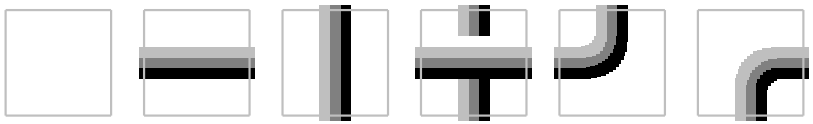
Example



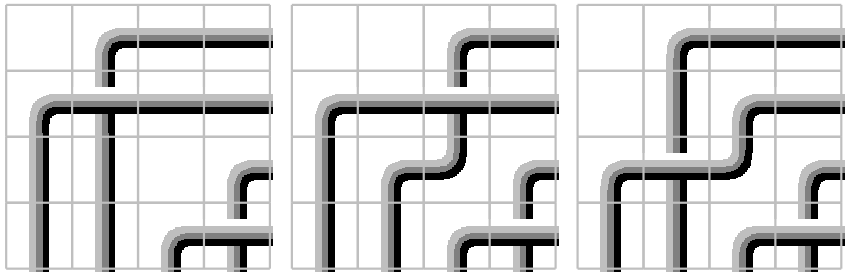
Our results

-less Pipe Dreams

A BPD of $w \in S_n$ is a tiling of $n \times n$ grid via



such that no pair of pipes intersects more than once BPDs can be used to compute double Schubert polynomials.

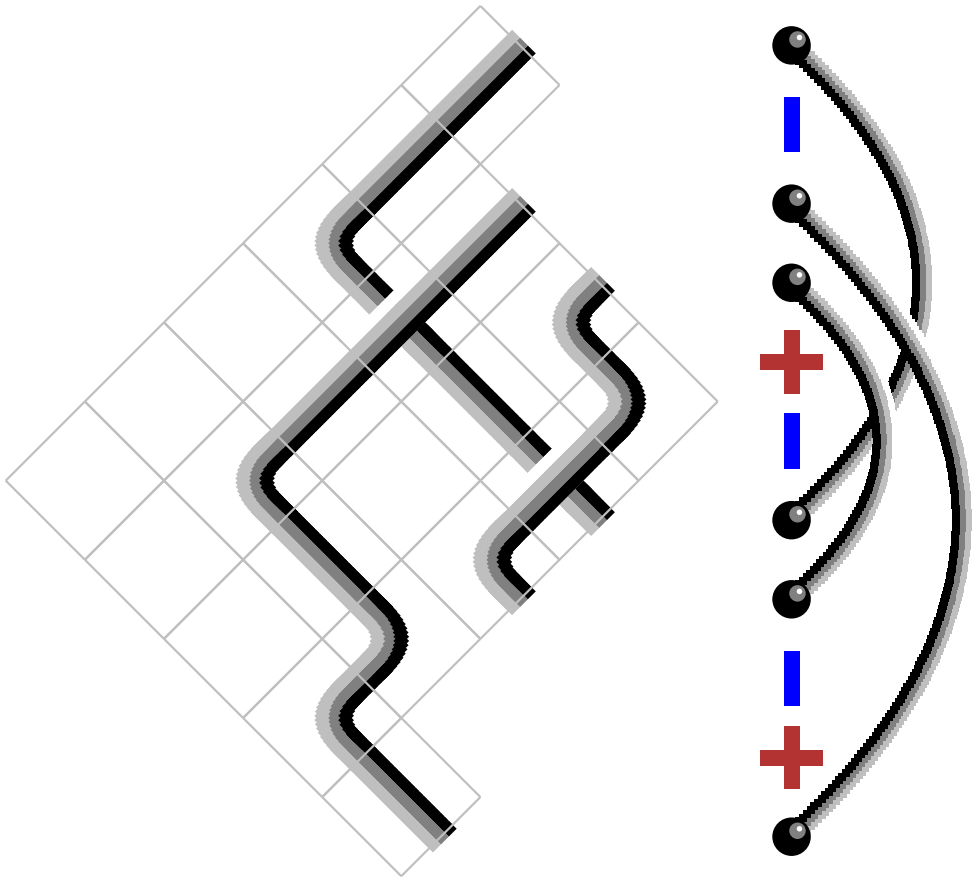


$$\begin{aligned}\mathfrak{S}_{2143} = &(x_1 - y_1)(x_3 - y_3) \\ &+ (x_1 - y_1)(x_1 - y_2) \\ &+ (x_1 - y_1)(x_2 - y_1).\end{aligned}$$

Schubert polynomials are polynomial representative of Schubert classes over flag variety.

Bumpless Pipe Dreams

The novel observation is that BPD fragments are naturally indexed by **clans**, rather than permutations.

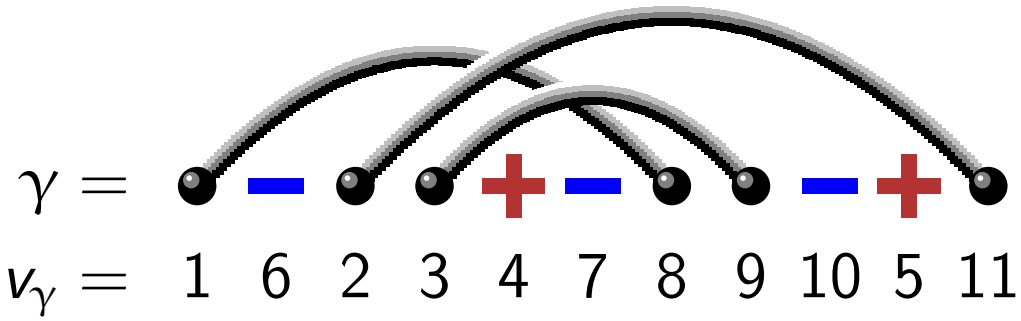


Main result — Localization

Let us denote the localization

$$\Upsilon_\gamma|_w = \begin{array}{c}\text{localization of fundamental}\\ \text{class of orbit closure at } wB/B.\end{array}$$

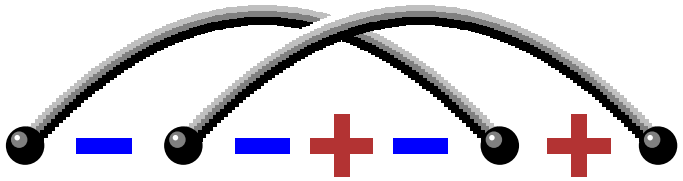
Our main result is a combinatorial formula of the localization at the minimal $w = v_\gamma$ such that $\Upsilon_\gamma|_w \neq 0$:



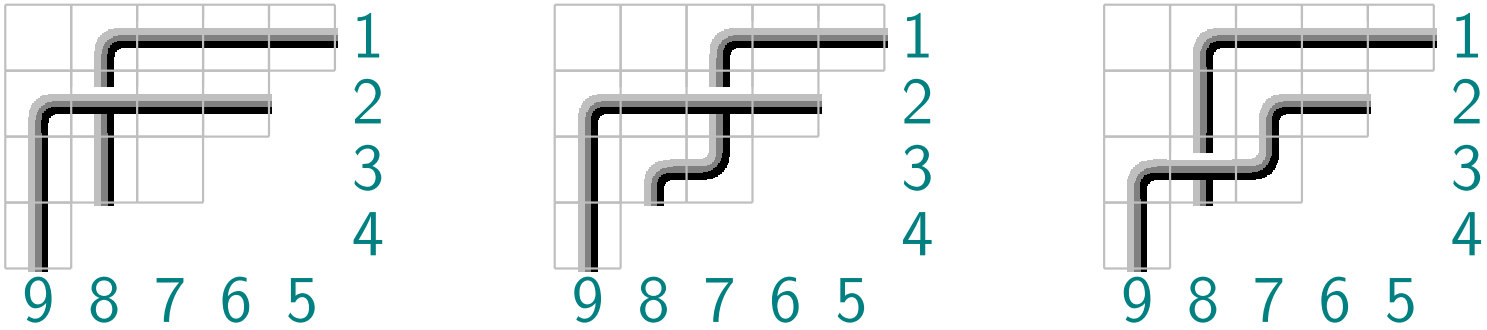
Theorem (Chen–Fan–X.–Yao, 2025+)

The localization $\Upsilon_\gamma|_{v_\gamma} \in \mathbb{Q}[y_1, \ldots, y_n]$ *is given by* **bumpless pipe dream fragments**.

Example

$\gamma =$ 

$v_\gamma = 1 \quad 5 \quad 2 \quad 6 \quad 3 \quad 7 \quad 8 \quad 4 \quad 9$



$$\begin{array}{ccccc} & & & & y_5\text{-}y_2 \\ & & & y_6\text{-}y_3 & y_5\text{-}y_3 \\ y_8\text{-}y_4 & y_7\text{-}y_4 & y_6\text{-}y_4 & y_5\text{-}y_4 & \end{array} \cdot \left(\begin{array}{ccccc} y_1\text{-}y_9 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & \\ 1 & 1 & y_3\text{-}y_7 & & \\ 1 & & & & \end{array} + \begin{array}{ccccc} y_1\text{-}y_9 & y_1\text{-}y_8 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & \\ 1 & 1 & 1 & & \\ 1 & & & & \end{array} + \begin{array}{ccccc} y_1\text{-}y_9 & 1 & 1 & 1 & 1 \\ y_2\text{-}y_9 & 1 & 1 & 1 & \\ 1 & 1 & 1 & & \\ 1 & & & & \end{array} \right)$$

Main result — Schubert expansion

We give a complete answer of Wyser and Yong. Assume

$$\Upsilon_\gamma(x;y) = \sum_{w \in S_n} c_{\gamma w}(y) \mathfrak{S}_w(x;y).$$

Theorem (Chen–Fan–X.–Yao, 2025+)

We have

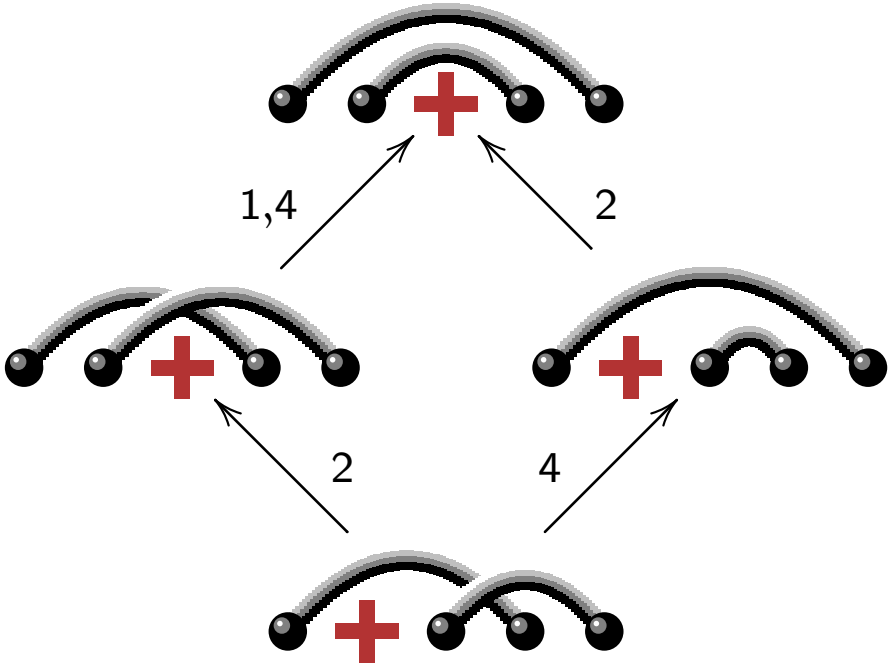
$$c_{\gamma w}(y) = \begin{matrix} \textit{Schubert polynomial} \\ \textit{of a partial permutation} \end{matrix} \textit{ or } 0$$

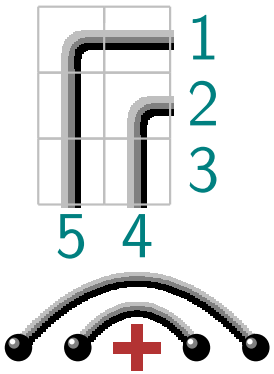
where the partial permutation is determined by $w * \gamma$.

Example

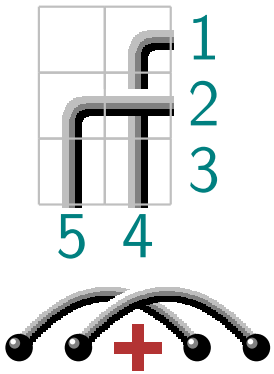
$\Upsilon_{\gamma=}$ (x,y)

$$= (y_1 - y_5)(y_2 - y_5)\mathfrak{S}_{\mathrm{id}}(x,y)$$
$$+ (y_1 - y_5)\mathfrak{S}_{s_2}(x,y)$$
$$+ (y_2 - y_4 + y_1 - y_5)\mathfrak{S}_{s_4}(x,y)$$
$$+ \mathfrak{S}_{s_1 s_2}(x,y)$$
$$+ \mathfrak{S}_{s_2 s_4 = s_4 s_2}(x,y).$$

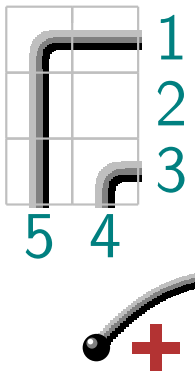


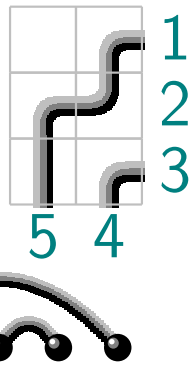


1

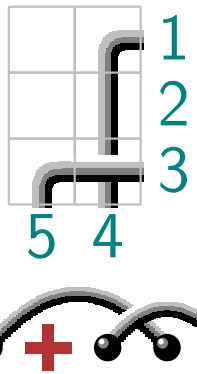


$y_1 - y_5$





$y_2 - y_4 + y_1 - y_5$



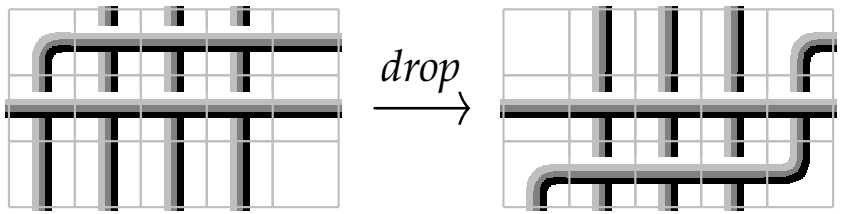
$(y_1 - y_5)(y_2 - y_5)$

Main result — Smoothness

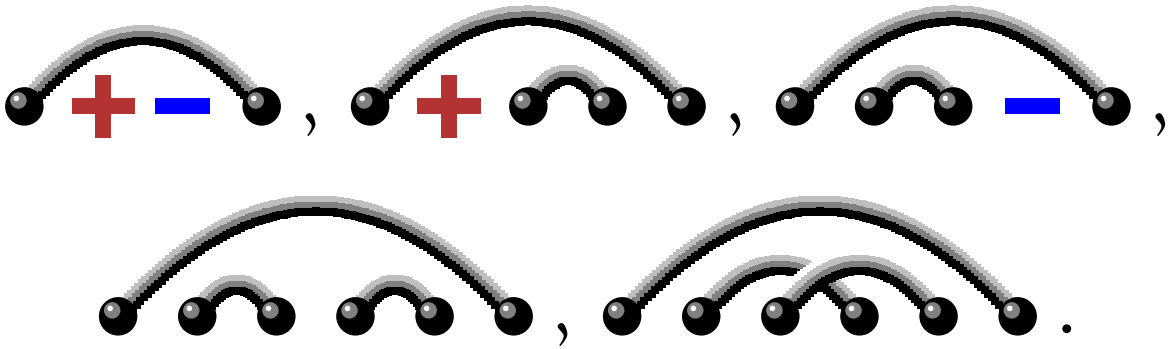
Corollary (Chen–Fan–X.–Yao, 2025+)

For a clan γ, the following statements are equivalent:

- *the orbit closure Y_γ is smooth at v_γ;*
- *the clan γ has exactly one BPD fragment, i.e. no space for*



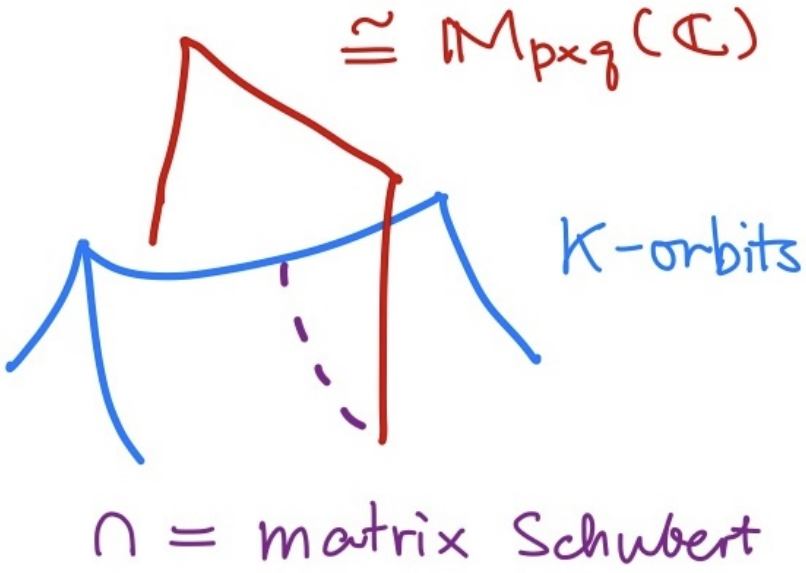
- *the clan γ avoids the following 5 patterns:*



Idea of the Proof

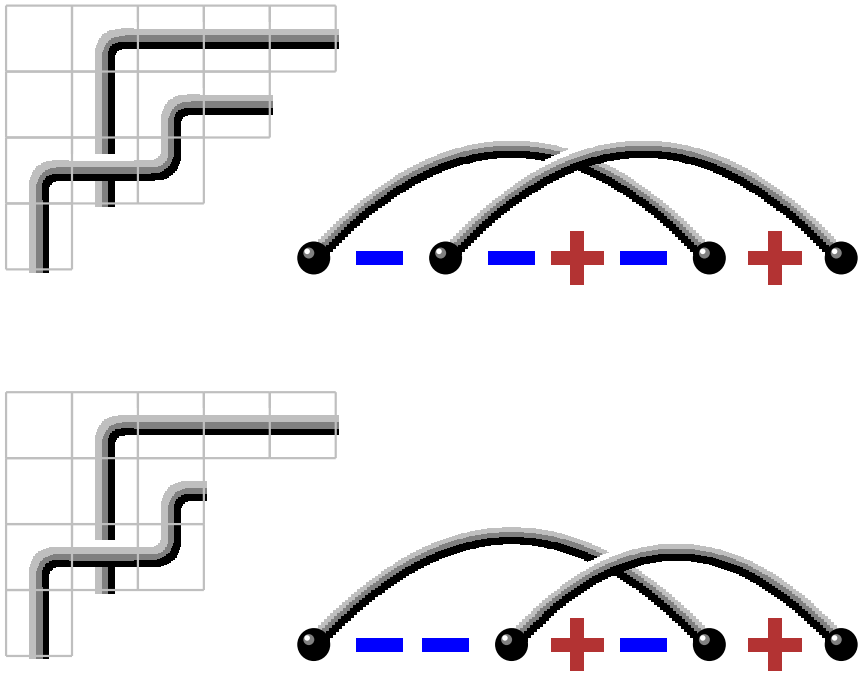
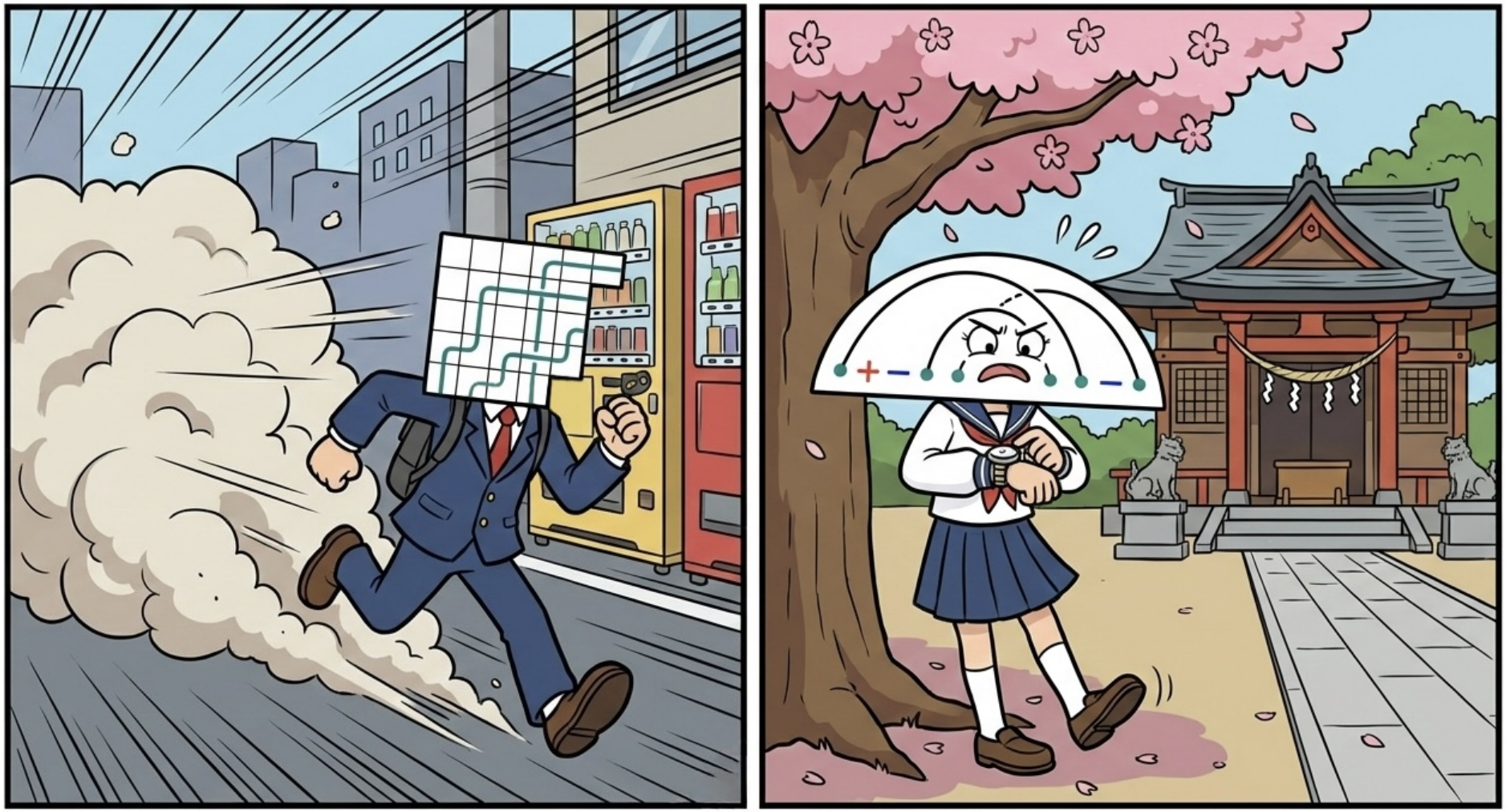
Geometric part of the proof

The initial case is by study the local structure of closed K-orbits at id.



Algebraic part of the proof

The inductive step is by study the local change of a clan.



Thank You!